

Payments & Lending Domain Knowledge Base (Vision- Optimised)

kb-L2-payments-lending-domain v1.0.0-slim

Optimised for: vision-generator, requirements-extractor,
epics-generator

Contains: PM implications (what to build), user journeys
(full), state machines (full), glossary (full)

Removed: field-level schemas, validation rules, API
implementation detail, regulatory full-text

kb-L2-payments-domain v1.0.0

Domain knowledge for UK consumer lending. All agents
creating epics, stories, or designs for lending products
MUST ground their decisions in this KB.

PD1: Lending Lifecycle

Customer Journey

1. **Discovery** — Customer encounters loan product via comparison site, direct marketing, or in-app promotion. FCA financial promotions rules (CONC 3) require representative APR and risk warnings in all ads.
2. **Eligibility Check** — Soft-search pre-qualification. No credit footprint. Returns indicative rate and limit. Must not constitute a credit agreement offer (CCA s.60).
3. **Application** — Full data capture (see PD2). FCA expects adequate explanations of the product (CONC 4.2.5R).
4. **KYC / Verification** — Identity verification (MLR 2017 reg.28), address verification, income verification. Electronic verification preferred; manual fallback required.

5. **Credit Decision** — Hard search at CRA, affordability assessment, automated or referred decision (see PD3).
6. **Offer** — Binding pre-contract credit information (SECCI) per CCA s.55A. Must include APR, total amount payable, repayment schedule, right to withdraw.
7. **Acceptance** — Customer accepts offer. 14-day withdrawal right begins (CCA s.66A).
8. **Disbursement** — Funds transferred via Faster Payments (typically < 2 hours) or BACS (next working day).
9. **Repayment** — Direct Debit collection via Bacs scheme. Continuous Payment Authority (CPA) restricted to two failed attempts for high-cost short-term credit (CONC 7.6.12R).
10. **Settlement / Closure** — Early settlement per CCA s.94. Settlement figure valid 28 days (CCA s.97). Account closed, CRA updated within 30 days.

Lender Journey

1. **Product Configuration** — Define rate bands, term ranges, fee structures, eligibility rules, scorecard thresholds.
2. **Lead Capture** — Ingest applications from direct channel, broker API, or aggregator feed.
3. **Application Processing** — Validate completeness, de-duplicate, enrich with CRA data.
4. **Identity Verification** — KYC checks against PEP/sanctions lists, document verification (passport/driving licence), liveness check where applicable.
5. **Credit Assessment** — CRA hard search, internal scorecard, policy rules engine.
6. **Affordability Check** — Income vs expenditure analysis, ONS benchmark comparison, stress test at +3pp interest rate.
7. **Offer Generation** — Risk-based pricing, fee calculation, APR computation (see PD4), SECCI document generation.
8. **Agreement Execution** — E-signature capture (eIDAS-compliant), agreement storage (6 years post-closure per FCA SYSC 9.1.1R).
9. **Funds Transfer** — Payment initiation via Faster Payments / BACS. Reconciliation against bank statement.
10. **Servicing** — Statement generation, payment processing, balance enquiries, complaints handling (DISP rules).
11. **Collections** — Arrears management per CONC 7.3 (treat customers fairly), forbearance options, income & expenditure review.
12. **Write-off / Closure** — Debt sale or write-off, CRA default registration (6 years), account closure.

Loan State Machine

State	Trigger	Validations	Next States	Timeout
Draft	Customer starts application	None	Submitted, Abandoned	30 days inactivity → Abandoned. Abandoned apps: data retained 90 days (customer can resume), then purged. Customer notified at 7 days and 21 days of inactivity.

State	Trigger	Validations	Next States	Timeout
Submitted	Customer completes & submits	All mandatory fields populated, consent captured	IdentityVerification	—
IdentityVerification	Submission accepted	KYC provider call, PEP/sanctions screen, document check	CreditCheck (pass), Referred (inconclusive), Closed (fail)	7 days → Closed
CreditCheck	Identity verified	CRA hard search, scorecard evaluation, policy rules	AffordabilityCheck (pass), Referred (marginal), Closed (decline)	—
AffordabilityCheck	Credit check passed	Income verification, expenditure analysis, stress test	Approved (pass), Referred (marginal), Closed (decline)	—
Referred	Any check inconclusive	Manual underwriter review, additional docs requested	Approved, Closed	14 days → Closed
Approved	All checks passed or underwriter approves	Final policy check, rate lock	OfferGenerated	24 hours → OfferGenerated (auto)
OfferGenerated	Approval confirmed	APR calculation, SECCI generation, fee schedule	OfferSent	—
OfferSent	Offer dispatched to customer	Delivery confirmation (email/SMS/in-app)	OfferAccepted, Closed (declined/expired)	30 days → Closed
OfferAccepted	Customer accepts offer	Acceptance timestamp, IP address, device fingerprint	AgreementSigned	7 days → Closed
AgreementSigned	E-signature captured	Signature validation, agreement PDF stored	Disbursing	—

State	Trigger	Validations	Next States	Timeout
Disbursing	Agreement executed	Bank account validation (CoP check), payment initiation	Active (funds confirmed), Disbursing (retry on failure)	3 retries → Referred
Active	Funds received by customer	Repayment schedule activated, DD mandate confirmed	InArrears, Settled, Closed	—
InArrears	Payment missed (DD failure + 3 day grace)	Arrears notification (CONC 7.3.4R), forbearance assessment	Active (payment received), Default (90+ days), Settled	—
Default	90+ days in arrears or 3 consecutive missed payments	Default notice issued (CCA s.87), CRA default registered	Settled (full payment), Closed (write-off/debt sale)	—
Settled	Full balance paid (including early settlement)	Settlement figure reconciliation, CRA update	Closed	—
Closed	Settlement confirmed, write-off, or application terminated	Final CRA update, record retention flag set	Terminal state	—

PM IMPLICATION

Every state transition is a notification trigger (email/SMS/push), a dashboard status update, and a compliance checkpoint. Epics must include notification templates, status tracking UI, and audit logging for each transition. Stories should map 1:1 to state transitions where possible. The 14-day withdrawal right (CCA s.66A) after acceptance requires a dedicated "withdrawal" flow that reverses disbursement — this is a separate epic.

PD2: Loan Origination

Application Data Model

Address History

Employment Types & Impact on Application

Income Verification by Type

Document Requirements by Applicant Type

Auto-Save

Joint Applications

Guarantor Loans

Top-Up Loans

PM IMPLICATION

Each employment type drives a different form path with conditional fields, different document upload requirements, and different income verification logic. Epics must include: (1) a dynamic form engine or branching logic per employment type, (2) document upload with OCR/classification, (3) Open Banking integration for auto-verification, (4) address lookup integration (PAF/OS Places), (5) auto-save with encryption. Stories should be sliced by employment type — each type is effectively a separate user journey through the same form. Joint applications, guarantor loans, and top-up loans are common product variants that each require distinct user journeys and should be planned as separate epics or feature sets.

PD3: Affordability & Creditworthiness

Creditworthiness Assessment (FCA CONC 5.2A)

Affordability Assessment (FCA CONC 5.2A.20R)

Open Banking Affordability

Debt Burden Ratio (DBR)

Responsible Lending Obligation

PM IMPLICATION

The application must collect granular expenditure data (not just a single "monthly outgoings" field) to enable ONS benchmark comparison. Stories must include: (1) expenditure breakdown form with ONS-aligned categories, (2) automated ONS benchmark flagging (if declared < benchmark, trigger manual review), (3) disposable income calculator shown to underwriters, (4) stress test results displayed alongside base-case affordability, (5) decline reason notification flow with CRA details. The affordability model is a separate microservice/module — it must be independently testable and auditable.

PD4: Pricing, Fees & APR

Interest Rate Types

APR Calculation

Fee Types & UK Regulatory Caps

Total Amount Payable

Representative Example (FCA-Required Format)

PM IMPLICATION

The offer screen is the most heavily regulated screen in the application. Every fee must be visible, the representative example must be displayed in the FCA-mandated format, and the total amount payable must be prominent. Epics must include: (1) APR calculation engine (iterative solver, independently auditable, tested against FCA worked examples), (2) offer screen with full representative example, (3) fee schedule display, (4) early repayment calculator (with correct cap applied based on remaining term), (5) rate change notification flow for variable-rate products. The APR engine is complex — it should be a separate, well-tested service. Stories should cover: offer screen layout, APR calculation accuracy tests, fee cap enforcement, early settlement quote generation, and representative example rendering across all channels (web, mobile, email, PDF).

PD5: Loan Offers & Agreements

Pre-Contract Information

PCCI (Pre-Contract Credit Information)

Credit Agreement

Right to Withdraw

Offer Expiry

Execution Process

PM IMPLICATION

The acceptance flow is legally prescribed — you cannot skip the PCCI, cannot hide the cooling-off right, and the e-signature must be legally valid. The withdrawal flow is often forgotten in product builds but is a legal requirement. Epics must include: (1) PCCI generation and display, (2) acceptance flow with e-signature capture, (3) agreement PDF generation and storage, (4) cooling-off period tracking, (5) withdrawal request flow (initiation, principal repayment, interest calculation, confirmation), (6) offer expiry management and re-application flow. Stories should cover: PCCI rendering across channels, e-signature capture and validation, agreement document generation, withdrawal happy path, withdrawal after disbursement (funds must be returned), offer expiry notification, and re-application after expiry.

PD6: Disbursement

Payment Methods

Faster Payments Detail

Timing: Immediate vs Cooling-Off

Account Verification

Failed Disbursement

Confirmation

PM IMPLICATION

Disbursement has critical edge cases: bank rejection, frozen account, withdrawal after disbursement (funds already sent, must be recovered). Stories must cover: (1) successful disbursement via FPS, (2) successful disbursement via BACS, (3) failed disbursement with retry, (4) failed disbursement with account update, (5) disbursement confirmation notifications, (6) cooling-off interaction (withdrawal before disbursement, withdrawal after disbursement with recovery flow), (7) CoP name mismatch handling. The disbursement status must be clearly visible in the customer dashboard and internal servicing tools.

PD7: Repayment & Collections

Repayment

Collections

Breathing Space (Debt Respite Scheme 2021)

PM IMPLICATION

Collections is heavily regulated with specific timelines and prescribed notices. The customer dashboard must show arrears status clearly. Notifications must follow the prescribed timeline and include mandatory content (debt advice signposting). Epics must include: (1) repayment schedule display and recalculation, (2) Direct Debit setup and management, (3) overpayment flow with term/payment choice, (4) arrears detection and automated reminder sequence, (5) formal arrears notice generation (CCA s.86B compliant), (6) default warning and registration flow, (7) forbearance request and management, (8) collections dashboard for internal teams. Stories must cover: normal repayment view, arrears dashboard, each stage of the collections timeline, forbearance request flow, debt advice signposting, CRA reporting, and default registration. Do not forget the Direct Debit Guarantee — customers can reclaim payments, and the system must handle indemnity claims.

PD8: Early Repayment & Settlement

Legal Right

Settlement Figure Calculation

Settlement Figure Validity

Compensation Caps

Partial Early Repayment

Settlement Process

PM IMPLICATION

Early settlement is a key differentiator for fintechs — zero fees should be prominent in marketing and the in-app experience. The settlement figure calculation must be penny-accurate and independently auditable. Epics must include:

- (1) settlement figure calculator (actuarial method, tested against worked examples), (2) settlement figure display (amount, validity date, payment instructions), (3) partial early repayment flow (lump sum input, term vs payment choice, schedule recalculation), (4) full settlement payment flow, (5) post-settlement confirmation and document generation, (6) CRA update on settlement. Stories must cover: settlement figure request and display, partial vs full settlement choice, payment execution, confirmation, CRA update, settlement letter generation, figure expiry and recalculation, and the edge case where a scheduled Direct Debit is collected after settlement (must be refunded).

PD9: Payment Processing

UK Payment Schemes

Faster Payments Detail

BACS Processing Cycle

Direct Debit Lifecycle

Payment Statuses

Reconciliation

Payment References

Sort Code Validation

Confirmation of Payee (CoP)

Open Banking Payments

PM IMPLICATION

Payment processing has multiple failure points and each must be handled gracefully. Stories must cover: (1) successful payment via each method (FPS, BACS, DD, card, Open Banking), (2) failed payment with specific reason codes (insufficient funds, account closed, invalid details), (3) Direct Debit setup flow (mandate creation, AUDDIS submission, confirmation), (4) Direct Debit cancellation by customer (ADDACS notification handling, alternative payment arrangement), (5) DD indemnity claim handling (refund to customer's bank, impact on loan balance, customer communication), (6) reconciliation discrepancy investigation and resolution, (7) Confirmation of Payee name mismatch

(warn customer, allow override or correction), (8) payment reference generation and validation, (9) suspense account management, (10) Open Banking payment initiation flow. The payment status must be visible in both customer-facing and internal dashboards, with clear indication of next steps for failed or returned payments.

PD10: PSD2 & Strong Customer Authentication (SCA)

PM IMPLICATION

SCA affects multiple user flows. Login can use biometric (one factor) for low-risk access, but loan acceptance and payment initiation need two factors. Stories must specify which SCA level each action requires. The dynamic linking requirement means the biometric prompt for loan acceptance must show the loan amount.

PD11: KYC & Identity Verification

PM IMPLICATION

KYC is not just "upload your ID". It's a multi-step process with multiple outcomes. Stories must cover: document selection, camera capture with quality guidance, selfie with liveness, OCR review, verification status tracking, retry on failure, manual review escalation, sanctions/PEP screening (even if automated, the result must be handled). The referred state is critical — what happens when auto-verification fails but the customer might still be legitimate?

PD12: FCA Consumer Duty

PM IMPLICATION

Consumer Duty affects EVERY screen. Product listing must not mislead. Offer screen must be crystal clear. Dashboard must make repayment easy. Early repayment must not be hidden. Complaints must be accessible. Stories should include acceptance criteria like: "a customer with no financial background can understand the total cost from the offer screen without external help".

PD13: Consumer Credit Act 1974 (as amended)

PM IMPLICATION

CCA compliance is not optional — non-compliance makes the loan unenforceable (the lender cannot recover the money). Every story that touches the agreement, offer, or repayment flow must have acceptance criteria that reference the relevant CCA section. The agreement generation feature must include ALL prescribed terms. The settlement figure feature must respond within 7 working days (s.97).

PD14: GDPR in Lending

PM IMPLICATION

GDPR affects the entire application flow. The privacy notice must be shown before data collection starts. Consent for marketing must be separate and optional (not bundled with T&Cs). The DSAR flow needs a story (even if it's a back-office process). Automated decision-making disclosure is required if using AI for credit scoring. Stories for data collection screens must specify the lawful basis in acceptance criteria.

PD15: Vulnerable Customers

PM IMPLICATION

Vulnerability is not a separate feature — it's a lens applied to every feature. Stories should include acceptance criteria like: "screen is usable with TalkBack at 200% text size", "if affordability check shows high debt-to-income ratio, system flags for manual review", "collections notifications use empathetic language and include debt advice contacts". Consider adding a "need help?" option on every screen that connects to human support.

PD16: Complaints & Dispute Resolution

PM IMPLICATION

Complaints handling needs its own epic or at minimum a set of stories. In-app: "Submit a complaint" flow, complaint status tracking, FOS information display. Back-office: complaint logging, assignment, investigation, response generation, FOS escalation tracking. Acceptance criteria must include the 5-day acknowledgment and 8-week resolution timelines.

PD17: Domain Glossary

Key terms used in UK consumer lending. Agents MUST use these terms consistently.

Term	Definition
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Term	Definition
APR	Annual Percentage Rate — the total cost of credit expressed as an annual percentage, including interest and mandatory fees. Calculated per CCA formula.
APRC	Annual Percentage Rate of Charge — used for mortgages (MCD), similar to APR but includes more cost components.
Arrears	When a borrower has missed one or more scheduled payments. Measured in months (1 month in arrears = 1 missed payment).
BACS	Bankers' Automated Clearing Services — UK payment system for Direct Debits and bank transfers. 3-day cycle.
CCJ	County Court Judgment — court order to repay a debt. Stays on credit file for 6 years.
CCA	Consumer Credit Act 1974 (as amended) — primary UK legislation governing consumer credit agreements.
CDD	Customer Due Diligence — identity verification required under Money Laundering Regulations.
CONC	Consumer Credit sourcebook — FCA's rules for consumer credit firms.
CoP	Confirmation of Payee — name-checking service that verifies account holder name before payment.
CRA	Credit Reference Agency — Experian, Equifax, or TransUnion. Holds credit file data.
DDI	Direct Debit Instruction — the mandate authorising a firm to collect payments from a customer's bank account.
Default	Formal notice that a borrower has failed to meet their obligations. Registered with CRAs, stays 6 years.
DISP	Dispute Resolution sourcebook — FCA's rules for complaints handling.
DMP	Debt Management Plan — informal arrangement to repay debts at reduced rate.
DSAR	Data Subject Access Request — GDPR right to obtain all personal data held by a firm.
EDD	Enhanced Due Diligence — additional identity checks for higher-risk customers (PEPs, high-risk countries).
FCA	Financial Conduct Authority — UK regulator for financial services firms.
Forbearance	Arrangements made with a borrower in financial difficulty (payment holiday, reduced payments, term extension).
FOS	Financial Ombudsman Service — independent body that resolves complaints between consumers and financial firms.
FPS	Faster Payments Service — UK real-time payment system. Funds arrive in seconds, 24/7.
Hard search	Credit check that leaves a visible footprint on the credit file. Other lenders can see it. Used for full applications.
IVA	Individual Voluntary Arrangement — formal agreement with creditors to repay debts over time.
KYC	Know Your Customer — the process of verifying a customer's identity and assessing risk.
MLR	Money Laundering Regulations 2017 — UK regulations implementing EU Anti-Money Laundering Directives.

Term	Definition
ONS	Office for National Statistics — provides expenditure benchmarks used in affordability assessments.
PCCI	Pre-Contract Credit Information — standardised document that must be provided before a credit agreement.
PEP	Politically Exposed Person — individual who holds or has held a prominent public function. Requires EDD.
PSD2	Payment Services Directive 2 — EU directive (retained in UK law) regulating payment services. Introduces SCA and Open Banking.
Representative APR	The APR that at least 51% of successful applicants will receive. Must be shown in advertising.
SCA	Strong Customer Authentication — two-factor authentication required by PSD2 for electronic payments and account access.
Settlement figure	The amount needed to repay a loan in full today. Includes principal, accrued interest, fees, minus any rebate.
Soft search	Credit check that does NOT leave a visible footprint. Used for eligibility checks and quotations.
Total amount payable	The total the borrower will pay over the life of the loan: principal + interest + all fees. Must be disclosed.
Total charge for credit	The total cost of credit to the borrower: interest + all mandatory fees. Used in APR calculation.
AISP	Account Information Service Provider — regulated entity that can access bank account data with customer consent under PSD2. Used for Open Banking affordability.
APP fraud	Authorised Push Payment fraud — customer is tricked into sending money to a fraudster.
Breathing space	Debt Respite Scheme (2021) — 60-day moratorium protecting debtors from creditor action. Interest and charges frozen.
CPA	Continuous Payment Authority — recurring card payment. Restricted to 2 failed attempts for high-cost credit (CONC 7.6.12R).
PISP	Payment Initiation Service Provider — regulated entity that can initiate payments from customer's bank account under PSD2.
SAR	Suspicious Activity Report — filed with National Crime Agency when fraud or money laundering is suspected.
VRP	Variable Recurring Payment — Open Banking payment with variable amounts within agreed parameters. Emerging capability.
AUDDIS	Automated Direct Debit Instruction Service — electronic system for setting up and cancelling Direct Debit mandates.
ADDACS	Automated Direct Debit Amendment and Cancellation Service — notifies originators of changes to or cancellations of DDIs.
ARUDD	Automated Return of Unpaid Direct Debits — notification that a Direct Debit collection has been returned unpaid.
Counter-offer	When a customer is declined for the requested amount but approved for a lower amount or different terms.
DBR	Debt Burden Ratio — total monthly debt obligations divided by net monthly income. Used to assess affordability.
I&E	Income and Expenditure review — detailed assessment of customer's financial situation for affordability and restructuring.

Term	Definition
Liability letter	Formal document stating outstanding loan balance, remaining term, and monthly payment.
Notice of Correction	Statement (up to 200 words) a customer can add to their credit file to explain circumstances.
Restructuring	Formal change to loan terms when forbearance is insufficient. Creates a new credit agreement.
Settlement letter	Document confirming loan fully repaid and account closed. Issued after settlement.
SFS	Standard Financial Statement — standardised format for income and expenditure reviews.

PD18: Product Configuration & Eligibility

Product Parameters

Eligibility Rules

Rate Tiering (Risk-Based Pricing)

Product Governance (FCA PROD rules)

PM IMPLICATION

Product configuration drives the entire eligibility and offer flow. Stories must cover: product catalog display (with representative APR and key facts), eligibility pre-check (soft search, instant result), rate personalisation (show personal rate after eligibility), product comparison (side-by-side for multiple products), and product governance reporting (arrears rate, complaints rate per product). The eligibility rules are the first filter — they determine what the customer sees before they even start an application.

PD19: Loan Servicing & Account Management

Statements

Account Changes

Interest Rate Changes (Variable Rate Products)

Annual Percentage Rate Statement

Liability Letters & Statements on Demand

Communication Preferences

PM IMPLICATION

Servicing is the longest phase of the loan lifecycle but often gets the least product attention. Stories must cover: annual statement generation and delivery, on-demand statement request, transaction history view, change of address/name/bank details flows (each with verification), rate change notification (variable products), communication preference management, and the distinction between marketing communications (opt-in) and regulatory communications (mandatory). The change-of-bank-details flow is particularly important — it's a common fraud vector and needs strong verification.

PD20: Fraud Prevention

Application Fraud Types

Fraud Detection Signals

Fraud Rules Engine

Authorised Push Payment (APP) Fraud

Regulatory Obligations

PM IMPLICATION

Fraud prevention must be embedded in the application flow, not bolted on. Stories must cover: device fingerprinting at app install, velocity checks during application, KYC liveness as anti-fraud (not just compliance), income verification cross-checks, fraud referral queue for operations team, SAR filing workflow (back-office), and the balance between fraud prevention and customer friction (too many checks = abandonment, too few = fraud losses). The fraud rules engine is a separate system that needs its own configuration stories.

PD21: Open Banking

Account Information Services (AISP)

Payment Initiation Services (PISP)

Variable Recurring Payments (VRP)

Customer Experience

PM IMPLICATION

Open Banking is a major differentiator for fintechs. Stories must cover: AISP consent flow (bank selection, redirect, return), transaction data display (categorised income/expenditure), affordability calculation using Open Banking data, PISP payment flow (for repayments and settlement), consent management (view active consents, revoke consent), re-authentication every 90 days, multi-bank aggregation, and fallback for customers who decline Open Banking (document upload path). The consent flow involves leaving the lender's app and returning — this must be seamless and handle all error cases (timeout, decline, bank app crash).

PD22: Financial Promotions

CONC 3: Financial Promotions

Required Content

Prominence Rules

Social Media & Digital

Record Keeping

PM IMPLICATION

Every screen that shows a loan product with any cost figure is a financial promotion. The product catalog, eligibility result, comparison page, and marketing emails all need representative APR and representative example. Stories must cover: product listing with compliant representative example, eligibility result display (personal rate + representative APR), marketing email templates with required content, push notification content compliance, and a financial promotions approval workflow (legal/compliance review before any promotion goes live).

PD23: Refunds & Adjustments

Refund Scenarios

Refund Methods

Regulatory Requirements

PM IMPLICATION

Refunds are often forgotten in product builds but are a regulatory requirement. Stories must cover: overpayment refund (automatic detection and processing), post-settlement DD refund (detect DD after settlement, auto-refund), compensation payment (manual trigger from complaints team), fee reversal (operations tool), and the edge case of refunding to a closed account. The refund status must be visible to the customer in their dashboard.

PD24: Credit Reporting Obligations

Monthly Reporting

What Triggers CRA Updates

Customer Disputes

PM IMPLICATION

Credit reporting is an ongoing operational obligation. Stories must cover: monthly CRA reporting batch job, real-time CRA updates for defaults and settlements, customer dispute handling workflow, and CRA data accuracy monitoring.

PD25: Loan Restructuring

Restructuring Options

Legal Requirements

Income & Expenditure Review

PM IMPLICATION

Restructuring needs: I&E collection form, restructuring option calculation, old vs new terms comparison display, new agreement signing flow, CRA update, and updated repayment schedule.